

Ossama Ahmed

AI AND AUTONOMOUS SYSTEM RESEARCH ASSOCIATE

Stahlwerkstraße 51, 44145 Dortmund, Germany

☎ (+49) 1782976484 | ✉ osa.ram2020@gmail.com | 📱 posquit0 | 🌐 www.linkedin.com/in/osama-ramadan

“Be the change that you want to see in the world.”

Research Interest

Bridging Classical Control and Machine Learning for Adaptive Robotic Systems

- **Core Objective:** Integrating data-driven learning methods with classical control and motion planning algorithms to develop autonomous systems that combine the adaptability of machine learning with the formal stability and safety guarantees of control theory.
- **Theoretical Methodology:** Leveraging information-theoretic approaches (e.g., Fisher information metrics and entropy parameters) to reason about model uncertainty, guide safe system identification, and maintain robust performance during time-varying model drifts.
- **Algorithmic Framework:** Designing modular, low-level orchestration architectures that embed physics-based kinematic models directly into learning loops, enhancing sample efficiency and closing the gap between virtual simulation and real-world deployment.

Education

Technical University of Hamburg (TUH)

Hamburg, Germany

MASTER'S OF SCIENCE IN MECHATRONICS ENGINEERING

2019 - 2023

- GPA: 1.6
- Thesis Topic: Predictive Path Planning with MPC for an Autonomous Forklift (STILL GmbH)

German University in Cairo (GUC)

Cairo, Egypt

BACHELOR OF SCIENCE IN MECHATRONICS ENGINEERING

2014 - 2019

- GPA: 0.94 (A+, with honors)
- Thesis Topic: Occupancy Grid Maps for Grounded Autonomous vehicles (UC3M)

Research Experience

Fraunhofer Institute for Material Flow and Logistics (IML)

Dortmund, Germany

RESEARCH ASSOCIATE (FULL-TIME)

Jun. 2023 - Present

- Led the research and development of automated system identification and optimization pipelines to systematically close the sim-to-real gap for diverse mobile robotic platforms (Ackermann, differential, and holonomic kinematics).
- Engineered a custom modular ROS2 hardware orchestration framework using Lifecycle nodes and low-level SocketCAN C++ drivers to control industrial motor state machines directly.
- Designed and validated high-fidelity rigid-body simulation models inside NVIDIA Isaac Sim using steady-state telemetry datasets and black-box optimization techniques.
- Co-authored 2 peer-reviewed papers published in flagship IEEE robotics conferences (ICRA 2024, CASE 2025) focusing on data-driven simulation modeling and automated parameter tuning.

STILL GmbH (in collaboration with TU Hamburg)

Hamburg, Germany

GRADUATE RESEARCH STUDENT & MASTER'S THESIS CANDIDATE

2022 - 2023

- Formulated and implemented a Predictive Path Planning pipeline utilizing Model Predictive Control (MPC) for an autonomous industrial forklift operating in constrained logistics environments.
- Developed a specialized research project course focused on fusing classical kinematics with predictive trajectory generation to optimize safety bounds and path smoothness.
- Evaluated algorithm performance across simulated edge-case scenarios, achieving major improvements in tracking accuracy and computational efficiency.

Universidad Carlos III de Madrid (UC3M)

Madrid, Spain

UNDERGRADUATE RESEARCH ASSISTANT (9-MONTH BACHELOR PROJECT)

2018 - 2019

- Conducted a 9-month fully immersive research project focused on environmental perception and mapping strategies for grounded autonomous vehicles.
- Implemented and optimized probabilistic Occupancy Grid Mapping algorithms to process sensor telemetry and handle high-uncertainty spatial data in real-time.
- Defended research findings as part of the formal Bachelor's Thesis requirement at GUC, receiving an outstanding grade evaluation.

Publications

Simulation Modeling of Highly Dynamic Omnidirectional Mobile Robots Based on Real-World Data

IEEE

INTERNATIONAL CONFERENCE ON ROBOTICS AND AUTOMATION (ICRA)

2024

- M. Wiedemann, **O. Ahmed**, A. Dieckhöfer, R. Gasoto, S. Kerner.
- Developed a data-driven pipeline utilizing Motion Capture systems to bridge the sim-to-real gap for holonomic mobile logistics platforms.

Automated Tuning of Non-Differentiable Rigid Body Simulation Models for Wheeled Mobile Robots

IEEE

IEEE INTERNATIONAL CONFERENCE ON AUTOMATION SCIENCE AND ENGINEERING (CASE)

2025

- M. Wiedemann, **O. Ahmed**, M. Hatwar, et al.
- Formulated a black-box optimization pipeline using real-world motion data to automatically tune non-differentiable physics simulation parameters across differential, Ackermann, and omnidirectional kinematics.

Skills

Robotics & Frameworks	ROS2 (Lifecycle Nodes, Composable Containers, Executors), NVIDIA Isaac Sim, URDF/Xacro, Gazebo
Programming & Software	C++ (17/20, OOP), Python, MATLAB / Simulink, Embedded Linux, Bash Scripting, Git, Docker
Control & Math Libraries	Eigen, NumPy, SciPy, PyTorch (Data-Driven Control, Numerical Optimization, System Identification)
Hardware & Protocols	PCB Design (EagleCAD), Embedded Systems, SocketCAN, CAN bus, SPI, I2C, Hardware-in-the-Loop (HIL) Testing
Languages	Araic (Native), English (C1), German (B2)

Extracurricular & Engineering Experience

Breeze Technologies

Hamburg, Germany

HARDWARE ENGINEER (WORKING STUDENT)

Feb. 2019 - 2023

- Designed, prototyped, and validated custom electronic PCB layouts for high-accuracy air quality environmental sensors.
- Developed and optimized low-level data acquisition and wireless transmission algorithms on microcontrollers for real-world IoT deployments.
- Formulated benchmarking frameworks to systematically evaluate sensor hardware performance and conducted comparative research on communication modules.

E-Ignition — Formula Student Driverless (TU Hamburg)

Hamburg, Germany

EMBEDDED & SELF-DRIVING SOFTWARE TEAM MEMBER

2019 - 2020

- Contributed to the software architecture design and low-level implementation of embedded systems driving the autonomous capabilities of the racing vehicle.
- Integrated real-time hardware-software constraints to ensure reliable execution of perception and localized motion control loops.

GUC Innovators — Shell Eco-Marathon Competition

Cairo, Egypt

HEAD OF ELECTRICAL SYSTEMS

2016 - 2019

- Led the electrical engineering team in designing and implementing the power train architecture and electrical control systems for an ultra-efficient urban concept vehicle.
- **Achievement:** Awarded **2nd Place Internationally** for the prestigious Innovation and Safety Award, validating the team's rigorous engineering and risk-mitigation designs.